Chemical Bonding: Detailed Midterm Exam Notes

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- Summary: Quick Reference Guide for Midterm

1. Quantum Numbers and Types

Quantum numbers are sets of numerical values that describe the unique quantum state of an electron in an atom. There are four types of quantum numbers that completely define an electron's position and behavior.

1.1 Principal Quantum Number (n)

The principal quantum number determines the main energy level or shell in which the electron resides. It is denoted by the symbol **n** and can take positive integer values: 1, 2, 3, 4, ... [1] [2] [3]

- **n = 1** corresponds to the K shell (closest to nucleus)
- **n = 2** corresponds to the L shell
- **n = 3** corresponds to the M shell
- **n = 4** corresponds to the N shell

The principal quantum number is related to the **size** of the orbital and the **energy** of the electron—higher values of n mean larger orbitals and higher energy levels [1] [4].

1.2 Azimuthal Quantum Number (l)

The azimuthal quantum number (also called the orbital angular momentum quantum number) determines the **shape** of the orbital and the subshell to which the electron belongs. It is denoted by **l** [1] [2] [3].

For a given value of n, l can take integer values from **0 to (n-1)** [1] [2]:

- When **l = 0**, it corresponds to an **s subshell** (spherical shape)
- When **l = 1**, it corresponds to a **p subshell** (dumbbell shape)
- When **l = 2**, it corresponds to a **d subshell** (complex shapes)
- When **l = 3**, it corresponds to an **f subshell** (more complex shapes)

Examples:

- For n = 1: l = 0 only (1s orbital)
- For n = 2: l = 0, 1 (2s, 2p orbitals)
- For n = 3: l = 0, 1, 2 (3s, 3p, 3d orbitals)
- For n = 4: l = 0, 1, 2, 3 (4s, 4p, 4d, 4f orbitals) [1] [2]

1.3 Magnetic Quantum Number (m or ml)

The magnetic quantum number determines the **orientation** of the orbital in three-dimensional space. It is denoted by **m** or **ml** [3] [5].

For a given value of l, m can take integer values from **–l to +l**, including zero [2] [5].

- For **s orbital (l = 0)**: m = 0 (1 orientation)
- For **p orbitals (l = 1)**: m = –1, 0, +1 (3 orientations: px, py, pz)
- For **d orbitals (l = 2)**: m = –2, –1, 0, +1, +2 (5 orientations)
- For **f orbitals (l = 3)**: m = –3, –2, –1, 0, +1, +2, +3 (7 orientations) [2] [5]

The total number of orientations for any subshell is given by **(2l + 1)** [attached_file:1].

1.4 Spin Quantum Number (s or ms)

The spin quantum number describes the intrinsic angular momentum (spin) of the electron. It is denoted by **s** or **ms** [3] [5].

The spin quantum number can have only two values [5]:

- **+½** (spin up, clockwise spin, represented by ↑)
- **–½** (spin down, anticlockwise spin, represented by ↓)

The spin quantum number is independent of the other three quantum numbers and determines the magnetic properties of atoms [5].

Summary Table

Quantum Number	Symbol	Meaning	Possible Values
Principal	n	Energy level, size	1, 2, 3, 4, ...
Azimuthal	l	Subshell, shape	0 to (n–1)
Magnetic	m or ml	Orientation	–l to +l
Spin	s or ms	Spin direction	+½ or –½

2. Principles for Electronic Configuration

Electronic configuration describes the distribution of electrons in atomic orbitals. Three fundamental principles govern how electrons fill orbitals.

2.1 Aufbau Principle

The Aufbau principle (from German "aufbauen" meaning "to build up") states that electrons fill atomic orbitals in order of **increasing energy**, starting from the lowest energy orbital [6] [7] [8].

Energy ordering of orbitals:
1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d ... [6] [7]

Key points:

- The (n+l) rule helps determine orbital filling order: orbitals with lower (n+l) values fill first
- If two orbitals have the same (n+l) value, the one with lower n fills first [attached_file:1]
- For example, 4s (n+l = 4+0 = 4) fills before 3d (n+l = 3+2 = 5)

2.2 Pauli Exclusion Principle

The Pauli exclusion principle states that **no two electrons in an atom can have the same set of all four quantum numbers** (n, l, m, ms) [6] [9] [8] [10].

Implications:

- Each orbital can hold a **maximum of two electrons**
- These two electrons must have **opposite spins** (+½ and –½)
- This ensures each electron has a unique quantum state [6] [9] [10]

Example: In the 1s orbital of helium:

- First electron: n=1, l=0, m=0, ms=+½
- Second electron: n=1, l=0, m=0, ms=–½

2.3 Hund's Rule

Hund's rule states that when filling degenerate orbitals (orbitals of equal energy), electrons occupy them singly with **parallel spins** before pairing up [6] [7] [8].

Key points:

- Electrons prefer to remain unpaired when possible
- All singly occupied orbitals in a subshell have electrons with the same spin direction
- Pairing only occurs after each orbital in the subshell has one electron [7] [8]

Example: For nitrogen (7 electrons): $1s^2\ 2s^2\ 2p^3$

- The three 2p electrons occupy three different p orbitals (px, py, pz) with parallel spins
- Configuration: 2p: ↑ ↑ ↑ (not ↑↓ ↑)

2.4 Electronic Configuration of Cations and Anions

For cations: Remove electrons starting from the outermost shell, typically in the order: p → s → d [9]

Example: Ca (Z=20): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$
Ca^{2+}: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (remove both 4s electrons)

For anions: Add electrons following the Aufbau principle [9]

Example: Cl (Z=17): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
Cl^-: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (add one electron to 3p)

3. Chemical Bonding

Chemical bonding is the attractive force that holds atoms or molecules together to form chemical compounds. Atoms form bonds to achieve a more **stable electronic configuration**, typically by attaining a noble gas configuration (octet rule) [11] [12][attached_file:1].

3.1 Cause of Chemical Bonding

Energy consideration:

- Bond formation is a **spontaneous process** because it leads to a **decrease in potential energy** of the system
- The lower the energy, the greater the stability
- Atoms bond together because the resulting molecule has lower energy than the separated atoms [attached_file:1]

Octet rule:

- Most atoms tend to gain, lose, or share electrons to achieve 8 electrons in their valence shell
- This configuration resembles that of noble gases, which are extremely stable [12][attached_file:1]

4. Energetics of Bond Formation

When two atoms approach each other to form a bond, several energy changes occur.

4.1 Potential Energy Curve

As atoms approach:

- **At large distances:** No interaction; potential energy is zero
- **As atoms approach:** Attractive forces dominate initially, causing potential energy to decrease
- **At equilibrium distance:** Potential energy reaches a minimum—this is the **bond length**
- **At very short distances:** Repulsive forces dominate due to electron-electron and nucleus-nucleus repulsion, causing potential energy to increase sharply [13] [14]

4.2 Bond Dissociation Energy (BDE)

Bond dissociation energy is the energy required to **break one mole of a particular bond** in the gas phase, converting the molecule into separate atoms [attached_file:1].

- BDE is always **positive** (endothermic process)
- Higher BDE indicates a **stronger bond**
- The energy released during bond formation equals the BDE [attached_file:1]

Example: $H_2 \rightarrow 2H$; BDE = 436 kJ/mol

5. Types of Chemical Bonds

There are three primary types of chemical bonds and one special type:

5.1 Ionic Bond

Formed by **complete transfer** of electrons from a metal to a nonmetal, creating oppositely charged ions [11] [12] [15]

5.2 Covalent Bond

Formed by **sharing** of electron pairs between nonmetal atoms [11] [12] [15]

5.3 Metallic Bond

Formed by **delocalization** of electrons among a lattice of metal atoms [11] [15] [16]

5.4 Coordinate (Dative) Bond

A special type of covalent bond where **both electrons** in the shared pair come from the **same atom** [attached_file:1] [17] [18] [19]

6. Ionic Bond and General Properties

6.1 Formation of Ionic Bond

An ionic bond forms when a metal atom **transfers** one or more valence electrons to a nonmetal atom [11] [12] [16].

Process:

1. Metal loses electrons → becomes **cation** (positive ion)
2. Nonmetal gains electrons → becomes **anion** (negative ion)
3. Oppositely charged ions attract → **ionic bond** forms [12] [16]

Example: Formation of NaCl

- Na ($1s^2\ 2s^2\ 2p^6\ 3s^1$) loses 1 electron → Na^+ ($1s^2\ 2s^2\ 2p^6$)
- Cl ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$) gains 1 electron → Cl^- ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$)
- $Na^+ + Cl^- \rightarrow NaCl$ (ionic compound) [12]

6.2 General Properties of Ionic Compounds

1. Physical state:

- Solid at room temperature
- Form crystalline lattice structures [11] [15]

2. Melting and boiling points:

- **High** melting and boiling points due to strong electrostatic forces between ions [11] [15] [20]

3. Electrical conductivity:

- **Do not conduct** electricity in solid state (ions fixed in lattice)
- **Conduct electricity** when molten or dissolved in water (ions become mobile) [11] [15] [20]

4. Solubility:

- Generally **soluble in polar solvents** (like water)
- **Insoluble in nonpolar solvents** (like benzene) [11] [15]

5. Brittleness:

- Ionic crystals are **brittle**—when force is applied, ions of like charge align and repel, causing the crystal to shatter [11]

6. Hardness:

- Generally **hard** due to strong ionic interactions [attached_file:1]

7. Energetics of Ionic Bond

The formation of an ionic compound from its elements involves several energy steps:

7.1 Key Energy Terms

1. Sublimation/Atomization energy (ΔHsub or ΔHat):

- Energy required to convert solid metal to gaseous atoms
- Always **endothermic** (positive value) [21] [22] [23]

2. Bond dissociation energy (D):

- Energy required to break bonds in diatomic nonmetal molecules (e.g., $Cl_2 \rightarrow 2Cl$)
- Always **endothermic** [21] [22]

3. Ionization energy (IE):

- Energy required to remove electron(s) from gaseous metal atoms
- Always **endothermic**
- For metals forming 2+ or 3+ ions, multiple ionization energies must be considered ($IE_1 + IE_2 + ...$) [21] [22] [23]

4. Electron affinity (EA):

- Energy released when electron(s) are added to gaseous nonmetal atoms
- Usually **exothermic** (negative value in most conventions)
- **Note:** Some sources define EA as positive, so sign convention must be carefully checked [21] [22]

5. Lattice energy (U or ΔHlatt):

- Energy released when gaseous ions combine to form one mole of solid ionic compound
- Always **exothermic** (large negative value)
- **Alternatively:** Energy required to break the ionic lattice into gaseous ions (endothermic definition) [21] [22]

6. Enthalpy of formation (ΔHf°):

- Energy change when one mole of compound forms from its elements in their standard states
- Can be exothermic or endothermic [21] [22] [23]

7.2 Relationship Between Energy Terms

Using **Hess's Law**, the enthalpy of formation can be related to all the individual energy steps:

$$\Delta H_f^\circ = \Delta H_{at}(\text{metal}) + \frac{1}{2}D(\text{X}_2) + \sum IE + \sum EA + U$$

Where:

- All endothermic processes have positive values
- All exothermic processes have negative values
- Sign conventions must be consistently applied [21] [23] [24]

8. Lattice Energy and Lattice Enthalpy

8.1 Definitions

Lattice energy (U):
The energy **released** when one mole of an ionic solid is formed from gaseous ions [21] [22] [25].

$$M^{n+}(g) + X^{n-}(g) \rightarrow MX(s) + \text{Energy (lattice energy)}$$

Lattice enthalpy (ΔHlatt):
The enthalpy change when one mole of an ionic solid is formed from (or broken into) gaseous ions. The sign depends on the direction of the process [21] [22] [25].

8.2 Factors Affecting Lattice Energy

1. Charge on ions:

- Higher ionic charges → **stronger** electrostatic attraction → **higher** lattice energy
- Lattice energy ∝ $\frac{q_+ \times q_-}{r_+ + r_-}$

2. Size of ions:

- Smaller ions → ions can get closer → **stronger** attraction → **higher** lattice energy
- Larger ions → **lower** lattice energy [attached_file:1]

Examples:

- NaCl has lower lattice energy than MgO (both charge and size effects)
- LiCl has higher lattice energy than NaCl (Li^+ is smaller than Na^+)

9. Born-Haber Cycle

The Born-Haber cycle is a thermochemical cycle that applies **Hess's Law** to calculate lattice energy, which cannot be measured directly [21] [22] [24] [26] [25].

9.1 Construction of Born-Haber Cycle

The cycle breaks down the formation of an ionic compound into individual steps:

For a general compound MX (where M = metal, X = halogen):

Step 1: Atomization/sublimation of metal

$$M(s) \rightarrow M(g) \quad \Delta H = \Delta H_{at} \quad (\text{endothermic})$$

Step 2: Dissociation of nonmetal molecule

$$\frac{1}{2}X_2(g) \rightarrow X(g) \quad \Delta H = \frac{1}{2}D \quad (\text{endothermic})$$

Step 3: Ionization of metal atom

$$M(g) \rightarrow M^+(g) + e^- \quad \Delta H = IE_1 \quad (\text{endothermic})$$

Step 4: Electron affinity of nonmetal

$$X(g) + e^- \rightarrow X^-(g) \quad \Delta H = EA \quad (\text{usually exothermic})$$

Step 5: Formation of ionic lattice

$$M^+(g) + X^-(g) \rightarrow MX(s) \quad \Delta H = U \quad (\text{exothermic})$$

Overall process:

$$M(s) + \frac{1}{2}X_2(g) \rightarrow MX(s) \quad \Delta H = \Delta H_f^\circ$$

9.2 Calculation Using Born-Haber Cycle

By Hess's Law:

$$\Delta H_f^\circ = \Delta H_{at} + \frac{1}{2}D + IE_1 + EA + U$$

Rearranging to solve for lattice energy:

$$U = \Delta H_f^\circ - \Delta H_{at} - \frac{1}{2}D - IE_1 - EA$$

Or more generally:

$$U = \Delta H_f^\circ - \sum(\text{all other enthalpy terms})$$

[21] [22] [23] [24] [25]

9.3 Example Calculation

Problem: Calculate the lattice energy of NaCl given:

- ΔHf°(NaCl) = –411 kJ/mol
- ΔHat(Na) = +108 kJ/mol
- D(Cl_2) = +242 kJ/mol (so ½D = +121 kJ/mol)
- IE_1(Na) = +496 kJ/mol
- EA(Cl) = –349 kJ/mol

Solution:

$$U = \Delta H_f^\circ - \Delta H_{at} - \frac{1}{2}D - IE_1 - EA$$

$$U = -411 - 108 - 121 - 496 - (-349)$$

$$U = -411 - 108 - 121 - 496 + 349$$

$$U = -787\ \text{kJ/mol}$$

The negative value indicates energy is **released** when the lattice forms (exothermic).

10. Solvation Energy and Solubility

10.1 Solvation Energy

Solvation energy is the energy **released** when gaseous ions dissolve in a solvent and become surrounded by solvent molecules [27] [28] [29] [30].

General process:

$$M^{n+}(g) + \text{solvent} \rightarrow M^{n+}(\text{solvated}) + \text{Energy}$$

Hydration energy is a specific case of solvation energy when the solvent is **water** [27] [28] [29] [30].

10.2 Factors Affecting Solvation Energy

1. Charge on ion:

- Higher charge → **stronger** ion-dipole interactions → **higher** solvation energy [27] [30]

2. Size of ion:

- Smaller ion → solvent molecules can approach closer → **stronger** interactions → **higher** solvation energy
- Larger ion → **lower** solvation energy [27] [30]

3. Polarity of solvent:

- More polar solvent → **stronger** ion-dipole interactions → **higher** solvation energy [attached_file:1] [29]

Example: Hydration energies decrease with increasing ionic size:

- Li^+: –520 kJ/mol
- Na^+: –405 kJ/mol
- Cs^+: –265 kJ/mol [28]

10.3 Solubility of Ionic Compounds

The solubility of an ionic compound depends on the **balance** between:

1. **Lattice energy** (energy needed to break the crystal lattice)
2. **Hydration energy** (energy released when ions are hydrated)

Enthalpy of solution:

$$\Delta H_{sol} = \Delta H_{lattice} + \Delta H_{hydration}$$

Where ΔHlattice is positive (endothermic—breaking lattice) and ΔHhydration is negative (exothermic—solvating ions) [27] [28] [31] [32].

For dissolution to occur favorably:

- If **|ΔHhydration| > ΔHlattice|**, then ΔHsol is **negative** (exothermic) → compound is **soluble** [27] [33]
- If **|ΔHhydration| < ΔHlattice|**, then ΔHsol is **positive** (endothermic) → compound is **less soluble** or **insoluble** [27] [33]

Note: Entropy also plays a role in solubility—dissolution increases entropy (disorder), which favors solubility even if ΔHsol is slightly positive [27] [29].

11. Covalent Bond and Its Modern Concept

11.1 Valence Bond Theory (VBT)

Valence Bond Theory, developed by **Linus Pauling**, explains covalent bonding through the **overlap of atomic orbitals** [34] [13] [35].

Key postulates of VBT:

1. Orbital overlap:

- A covalent bond forms when **half-filled atomic orbitals** from two atoms overlap [34] [13] [35]
- The overlapping region has high electron density, which attracts both nuclei and holds them together [34] [35]

2. Directional nature:

- Covalent bonds are **directional**—they form in the direction of maximum orbital overlap [34] [35]
- This explains molecular geometry and bond angles [34]

3. Number of bonds:

- The number of covalent bonds an atom can form equals the number of **unpaired electrons** in its valence shell [34] [35]

4. Types of bonds:

- **Sigma (σ) bonds:** formed by head-on (axial) overlap
- **Pi (π) bonds:** formed by lateral (side-by-side) overlap [34] [13] [36] [35]

12. Orbital Overlapping and Its Types

12.1 Types of Orbital Overlap

A. Sigma (σ) Bond:

A sigma bond is formed by **head-on or axial overlap** of atomic orbitals along the internuclear axis [13] [37] [36] [35] [14].

Types of σ overlaps:

1. s-s overlap:

- Overlap between two s orbitals
- Example: H–H bond in H_2 molecule [35] [14]

2. s-p overlap:

- Overlap between an s orbital and a p orbital
- Example: H–Cl bond in HCl molecule [13] [35]

3. p-p overlap (head-on):

- Overlap between two p orbitals along their axes
- Example: Cl–Cl bond in Cl_2 molecule [13] [35] [14]

Characteristics of σ bonds:

- **Cylindrically symmetrical** about the internuclear axis
- **Stronger** than π bonds due to greater overlap
- Allow **free rotation** around the bond axis
- Electron density is **concentrated** between the two nuclei [13] [37] [36]

B. Pi (π) Bond:

A pi bond is formed by **lateral or sidewise overlap** of parallel p orbitals [13] [37] [36] [35].

Type of π overlap:

- **p-p overlap (lateral):**
  - Two parallel p orbitals overlap sideways
  - Forms two regions of electron density, one above and one below the internuclear axis
  - Example: Second and third bonds in C=C (double bond) and C≡C (triple bond) [37] [36]

Characteristics of π bonds:

- **Weaker** than σ bonds due to less effective overlap
- Electron density is **above and below** the internuclear axis (not along it)
- **Restricts rotation** around the bond axis
- A π bond **cannot exist independently**—it always accompanies a σ bond [37] [36] [35]

12.2 Comparison of Sigma and Pi Bonds

Property	Sigma (σ) Bond	Pi (π) Bond
Type of overlap	Head-on/Axial	Lateral/Sidewise
Orbitals involved	s-s, s-p, p-p (axial)	p-p (parallel)
Strength	Stronger	Weaker
Electron density	Along internuclear axis	Above and below axis
Rotation	Free rotation possible	Restricted rotation
Existence	Can exist independently	Always with σ bond

13. Molecular Orbital Theory (MOT)

Molecular Orbital Theory provides a more advanced description of bonding that overcomes some limitations of VBT, particularly for molecules with delocalized electrons [38] [39] [40] [41].

13.1 Basic Principles of MOT

1. Formation of molecular orbitals (MOs):

- Atomic orbitals (AOs) from combining atoms merge to form **molecular orbitals** [38] [39] [40]
- The number of MOs formed equals the number of AOs combined [attached_file:1] [38]

2. Linear combination of atomic orbitals (LCAO):

- MOs are formed by the **linear combination** of AOs of comparable energy [39] [40]

3. Types of molecular orbitals:

a) Bonding molecular orbitals (σ, π):

- Formed by **constructive (in-phase) interference** of atomic orbitals
- Have **lower energy** than the original atomic orbitals
- Electron density is **concentrated between** the nuclei, stabilizing the molecule

- **Increase stability** when occupied [38] [39] [40] [41] [42]

*b) Antibonding molecular orbitals (σ, π):**

- Formed by **destructive (out-of-phase) interference** of atomic orbitals
- Have **higher energy** than the original atomic orbitals
- Electron density is **depleted between** the nuclei (node present)
- **Decrease stability** when occupied
- Denoted with an asterisk (*) [38] [39] [40] [41] [42]

c) Nonbonding molecular orbitals:

- Have the **same energy** as the original atomic orbitals
- Do not contribute to bonding or antibonding [38] [40]

13.2 Energy Level Ordering

For homonuclear diatomic molecules (e.g., O_2, F_2, N_2):

The general order of molecular orbitals depends on the element:

For O_2 and F_2 ($Z \geq 8$):

For Li_2 through N_2 ($Z < 8$):

The difference is due to **sp mixing** in lighter elements [attached_file:1] [43].

13.3 Filling Molecular Orbitals

Electrons fill molecular orbitals following:

- **Aufbau principle:** Fill lowest energy MOs first
- **Pauli exclusion principle:** Maximum 2 electrons per MO with opposite spins
- **Hund's rule:** In degenerate MOs, electrons occupy singly with parallel spins before pairing [attached_file:1] [40]

14. Formation of Molecular Orbitals

14.1 Molecular Orbital Diagrams

MO diagrams show:

- Energy levels of atomic orbitals on the sides
- Energy levels of molecular orbitals in the center
- Electron occupancy represented by arrows [38] [40] [41]

Example: H_2 molecule

- Two 1s atomic orbitals combine

- Form σ_{1s} (bonding) and σ^*_{1s} (antibonding) MOs
- Two electrons occupy σ_{1s} (lower energy)
- σ^*_{1s} remains empty
- Result: stable H_2 molecule [41] [42] [43]

Example: He_2 molecule

- Two 1s atomic orbitals from each He atom combine
- Four electrons: two in σ_{1s}, two in σ^*_{1s}
- Bond order = 0 (see next section)
- Result: He_2 does not exist [40] [43]

14.2 Heteronuclear Diatomic Molecules

For molecules like CO or HF, where atoms have different electronegativities:

- Atomic orbitals have **different energies**
- The more electronegative atom has lower-energy orbitals
- MOs are **unsymmetrically distributed**—more electron density near the more electronegative atom [attached_file:1] [41]

15. Energy Level Diagram of Molecular Orbitals

Energy level diagrams visually represent:

1. Relative energies of atomic orbitals (left and right sides)
2. Molecular orbitals formed (center)
3. Electron distribution in the molecule

Key features:

- **Horizontal lines** represent energy levels
- **Arrows (↑↓)** represent electrons with their spins
- **Dashed lines** connect AOs to the MOs they form
- Energy increases vertically [38] [40] [41]

16. Electronic Configuration and Molecular Behavior and Properties of Covalent Bond

16.1 Bond Order

Bond order is a measure of the **number of chemical bonds** between a pair of atoms [38] [41] [attached_file:1].

Formula:

$$\text{Bond Order (BO)} = \frac{1}{2}(N_b - N_a)$$

Where:

- N_b = number of electrons in bonding molecular orbitals
- N_a = number of electrons in antibonding molecular orbitals [38] [41] [attached_file:1]

Interpretation:

- **BO = 0:** No bond forms (molecule does not exist)
- **BO = 1:** Single bond
- **BO = 2:** Double bond
- **BO = 3:** Triple bond
- **Higher BO** → stronger bond, shorter bond length, higher bond dissociation energy [38] [41] [attached_file:1]

Examples:

- **H_2:** $\frac{1}{2}(2-0) = 1$ → single bond
- **O_2:** $\frac{1}{2}(10-6) = 2$ → double bond
- **N_2:** $\frac{1}{2}(10-4) = 3$ → triple bond
- **He_2:** $\frac{1}{2}(2-2) = 0$ → no bond (does not exist) [43]

16.2 Magnetic Properties

Diamagnetic:

- All electrons are **paired**
- Weakly repelled by a magnetic field
- Example: N_2, F_2 [38] [40]

Paramagnetic:

- Contains **unpaired electrons**
- Attracted to a magnetic field
- Example: O_2 (has 2 unpaired electrons in π^*_2p orbitals) [38] [40] [43]

MOT successfully explains the **paramagnetism of O_2**, which VBT cannot predict from the Lewis structure [40] [43].

16.3 Bond Length and Stability

- **Higher bond order → shorter bond length → greater stability**
- **Lower bond order → longer bond length → lesser stability** [38] [41]

Example: Bond lengths in diatomic molecules

- N_2 (BO = 3): 110 pm
- O_2 (BO = 2): 121 pm
- F_2 (BO = 1): 142 pm

17. Coordinate or Dative Bond

A **coordinate bond** (also called dative bond) is a type of covalent bond where **both electrons in the bonding pair are donated by one atom** [17] [18] [19] [44] [45][attached_file:1].

17.1 Conditions for Formation

For a coordinate bond to form:

1. One atom must have a **lone pair of electrons** (donor)
2. The other atom must have a **vacant low-lying orbital** (acceptor) [17] [18] [19] [45]

17.2 Representation

Coordinate bonds are represented by an **arrow (→)** pointing from the **donor to the acceptor** [17] [18] [19].

Example: N → H^+ in NH_4^+

17.3 Examples of Coordinate Bonds

1. Ammonium ion (NH_4^+):

- Ammonia (NH_3) has a lone pair on nitrogen
- H^+ ion has an empty 1s orbital
- Nitrogen donates its lone pair to H^+
- **$NH_3 + H^+ \rightarrow NH_4^+$** [17] [18] [19] [44][^102]

2. Hydronium ion (H_3O^+):

- Water (H_2O) has lone pairs on oxygen
- H^+ has an empty 1s orbital
- Oxygen donates a lone pair to H^+
- **$H_2O + H^+ \rightarrow H_3O^+$** [44][^102]

3. Ammonia-boron trifluoride adduct ($NH_3 \rightarrow BF_3$):

- NH_3 has a lone pair on nitrogen (donor)
- BF_3 has an empty orbital on boron (acceptor)
- **$NH_3 + BF_3 \rightarrow NH_3BF_3$** [44] [45][^102]

4. Aluminum chloride dimer (Al_2Cl_6):

- At low temperatures, two $AlCl_3$ molecules dimerize

- Chlorine atoms donate lone pairs to aluminum atoms
- Forms two coordinate bonds in the dimer structure [17] [18] [46]

5. Carbon monoxide (CO):

- One of the bonds between C and O is a coordinate bond where carbon donates both electrons to oxygen [19] [44]

17.4 Properties of Coordinate Compounds

- Once formed, a coordinate bond is **indistinguishable** from a normal covalent bond [19] [45]
- Coordinate bonds have **directional** character [^102]
- Compounds with coordinate bonds have **lower melting and boiling points** than ionic compounds [^102]
- Coordinate bonds are **weaker than ionic bonds** but comparable to covalent bonds [^102]

18. Partial Covalent Character of Ionic Bond: Fajan's Rules

Although we classify bonds as ionic or covalent, most bonds have characteristics of both. **Fajan's rules** predict when an ionic bond will have significant **covalent character** [47] [48] [49] [50] [51] [52].

18.1 Statement of Fajan's Rules

An ionic bond exhibits more covalent character when:

1. The **cation is small** with **high charge**
2. The **anion is large** with **high charge**
3. The cation has a **pseudo-noble gas configuration** (18-electron outer shell) rather than a noble gas configuration (8-electron outer shell) [47] [48] [49] [50] [52]

18.2 Explanation: Polarization

Polarization is the distortion of the electron cloud of an anion by a cation [47] [48] [49] [50] [51].

Polarizing power (of cation):

- Ability of a cation to distort the electron cloud of an anion
- Increases with: **smaller size**, **higher charge** [47] [48] [50] [51]

Polarizability (of anion):

- Ease with which an anion's electron cloud can be distorted
- Increases with: **larger size**, **higher charge** [47] [48] [49] [50] [51]

When polarization is **high**, the electron cloud is significantly displaced toward the cation, creating shared electron density between the ions—this resembles **covalent bonding** [47] [48] [49] [51].

18.3 Detailed Rules

Rule 1: Effect of cation size

- **Smaller cation** → higher charge density → greater polarizing power → more covalent character
- **Larger cation** → lower charge density → less polarizing power → more ionic character

Example: LiCl > NaCl > KCl > RbCl > CsCl (decreasing covalent character) [50]

Rule 2: Effect of anion size

- **Larger anion** → more easily polarized → more covalent character
- **Smaller anion** → less easily polarized → more ionic character

Example: NaI > NaBr > NaCl > NaF (decreasing covalent character) [49] [50]

Rule 3: Effect of ionic charge

- **Higher charge** on cation or anion → greater polarization → more covalent character

Example: $AlCl_3$ > $MgCl_2$ > NaCl (decreasing covalent character) [48] [49]

Rule 4: Electronic configuration of cation

- Cations with **pseudo-noble gas configuration** (d^{10}) have greater polarizing power than those with **noble gas configuration** (s^2p^6)

Example:

- Ag^+ (pseudo-noble: [Kr]$4d^{10}$) is more polarizing than Na^+ (noble gas: [Ne])
- AgCl has more covalent character than NaCl, even though Ag^+ is larger [50] [52]

18.4 Consequences of Covalent Character

Ionic compounds with high covalent character show:

1. **Lower melting and boiling points** than expected for purely ionic compounds [50]
2. **Solubility in nonpolar solvents** (in addition to polar solvents) [50]
3. **Greater thermal instability** (easier decomposition) [50]

Example: AlI_3 has such high covalent character that it's soluble in organic solvents and has a low melting point.

19. Polarity of Covalent Bond

19.1 Polar vs. Nonpolar Covalent Bonds

Nonpolar covalent bond:

- Formed between **atoms of the same element** or atoms with **equal electronegativity**
- Electron pair is **equally shared**

- **No charge separation**
- Examples: H_2, Cl_2, O_2, N_2 [53] [54] [55] [56]

Polar covalent bond:

- Formed between atoms with **different electronegativities**
- Electron pair is **unequally shared**—pulled toward the more electronegative atom
- Creates **partial charges**: δ^+ (delta positive) on less electronegative atom, δ^- (delta negative) on more electronegative atom
- Examples: H–Cl, C–O, N–H [53] [54] [55] [56]

19.2 Factors Determining Bond Polarity

Electronegativity difference (Δχ):

- The greater the difference in electronegativity, the more polar the bond [53] [54] [55]

General guidelines:

- **Δχ < 0.4:** Nonpolar covalent bond
- **0.4 < Δχ < 1.7:** Polar covalent bond
- **Δχ > 1.7:** Ionic bond (approximate) [55]

Example: In HCl

- $\chi(Cl) = 3.0$, $\chi(H) = 2.1$
- $\Delta\chi = 0.9 \rightarrow$ polar covalent bond
- Electron density is greater around Cl (δ^-) than H (δ^+) [53] [54] [55]

20. Electronegativity

Electronegativity (χ) is the ability of an atom to **attract bonding electrons** toward itself in a covalent bond [53] [54] [55] [56].

20.1 Trends in Electronegativity

Across a period (left to right):

- Electronegativity **increases**
- Reason: increasing nuclear charge, similar shielding [53] [55]

Down a group (top to bottom):

- Electronegativity **decreases**
- Reason: increasing atomic size, more shielding of valence electrons [53] [55]

Most electronegative elements:

- **Fluorine (F)** is the most electronegative element ($\chi = 4.0$)

- Order: F > O > N > Cl [53] [55]

Least electronegative elements:

- Alkali metals (Li, Na, K, etc.) are the least electronegative [55]

20.2 Pauling Scale

The most commonly used scale for electronegativity, developed by Linus Pauling, assigns values from approximately 0.7 (Cs) to 4.0 (F) [53] [55].

21. Dipole Moment and Geometry of Molecules

21.1 Dipole Moment

Dipole moment (μ) is a measure of the polarity of a bond or molecule [53] [54] [57] [attached_file:1].

Formula:

$$\mu = q \times d$$

Where:

- μ = dipole moment
- q = magnitude of partial charge
- d = distance between charges (bond length)

Units:

- Measured in **Debye (D)** or **Coulomb-meters (C·m)**
- 1 D = 3.34 × 10^{-30} C·m [53] [54]

Direction:

- Dipole moment is a **vector quantity**
- Convention: arrow points from δ^+ to δ^- (toward more electronegative atom) [53] [54] [55]

21.2 Percent Ionic Character

The observed dipole moment can be used to calculate the **percent ionic character** of a bond:

$$\%\ \text{ionic character} = \left(\frac{\mu_{\text{observed}}}{\mu_{\text{ionic}}} \right) \times 100$$

Where:

- μobserved = measured dipole moment
- μionic = dipole moment if the bond were 100% ionic (calculated as $e \times d$) [attached_file:1]

Example: For HCl

- Bond length (d) = 127 pm = 1.27 × 10^{-10} m

- $\mu_{ionic} = (1.6 \times 10^{-19}\ C)(1.27 \times 10^{-10}\ m) = 2.03 \times 10^{-29}\ C \cdot m$
- $\mu_{observed} = 3.44 \times 10^{-30}\ C \cdot m$
- % ionic character = (3.44/20.3) × 100 ≈ 17% [attached_file:1]

21.3 Molecular Polarity and Geometry

For **molecules with more than two atoms**, overall polarity depends on:

1. **Polarity of individual bonds**
2. **Molecular geometry** (arrangement of bonds in 3D space) [53] [54] [55] [57]

Vector addition of bond dipoles determines net molecular dipole:

Nonpolar molecules with polar bonds:

- If bond dipoles **cancel due to symmetry**, the molecule is **nonpolar**
- Examples: CO_2 (linear), CCl_4 (tetrahedral), BF_3 (trigonal planar) [53] [55] [57]

Polar molecules:

- If bond dipoles **do not cancel**, the molecule is **polar**
- Examples: H_2O (bent), NH_3 (trigonal pyramidal), $CHCl_3$ (tetrahedral but asymmetric) [53] [54] [55] [57]

Key point: Symmetrical molecules are generally nonpolar; **asymmetrical molecules** are generally polar [53] [57].

22. Hybridization: Characterization and Types

Hybridization is the concept of **mixing atomic orbitals** of similar energy to form new **hybrid orbitals** that are equivalent in energy and shape, optimized for bonding [58] [59] [60] [61] [62] [63].

22.1 Why Hybridization?

- Explains observed **molecular geometries** and **bond angles**
- Accounts for the formation of **equivalent bonds** (e.g., four identical C–H bonds in methane)
- Predicts the **directional nature** of covalent bonds [58] [59] [62]

22.2 Types of Hybridization

A. sp Hybridization

Mixing: 1 s orbital + 1 p orbital → **2 sp hybrid orbitals** [58] [59] [60] [61]

Characteristics:

- **Linear geometry**
- **Bond angle:** 180°
- **% character:** 50% s, 50% p

- **Unhybridized orbitals:** 2 p orbitals remain (available for π bonding) [59] [60]

Examples:

- $BeCl_2$, BeH_2
- C_2H_2 (acetylene): each carbon is sp hybridized [58] [59] [60]

B. sp^2 Hybridization

Mixing: 1 s orbital + 2 p orbitals → **3 sp^2 hybrid orbitals** [58] [59] [60] [61] [63]

Characteristics:

- **Trigonal planar geometry**
- **Bond angle:** 120°
- **% character:** 33% s, 67% p
- **Unhybridized orbitals:** 1 p orbital remains (available for π bonding) [59] [60] [63]

Examples:

- BF_3, BCl_3
- C_2H_4 (ethene): each carbon is sp^2 hybridized
- Carbonyl carbon in aldehydes and ketones [58] [59] [61] [63]

C. sp^3 Hybridization

Mixing: 1 s orbital + 3 p orbitals → **4 sp^3 hybrid orbitals** [58] [59] [60] [61] [62] [63]

Characteristics:

- **Tetrahedral geometry**
- **Bond angle:** 109.5° (109°28')
- **% character:** 25% s, 75% p
- **Unhybridized orbitals:** None [59] [60] [61]

Examples:

- CH_4 (methane)
- NH_3 (ammonia)—with one lone pair
- H_2O (water)—with two lone pairs
- Diamond (each C is sp^3) [58] [59] [62] [63]

D. sp^3d Hybridization

Mixing: 1 s orbital + 3 p orbitals + 1 d orbital → **5 sp^3d hybrid orbitals** [58] [59] [60] [61]

Characteristics:

- **Trigonal bipyramidal geometry**
- **Bond angles:** 90° (axial-equatorial), 120° (equatorial-equatorial), 180° (axial-axial)

- **Positions:** 3 equatorial (in a plane), 2 axial (above and below the plane) [59] [61]

Examples:

- PCl_5 (phosphorus pentachloride)
- SF_4 (with one lone pair)—seesaw shape [attached_file:1]

E. sp^3d^2 Hybridization

Mixing: 1 s orbital + 3 p orbitals + 2 d orbitals → **6 sp^3d^2 hybrid orbitals** [58] [59] [60] [61]

Characteristics:

- **Octahedral geometry**
- **Bond angle:** 90°, 180°
- All positions are equivalent [59] [61]

Examples:

- SF_6 (sulfur hexafluoride)
- XeF_4 (with two lone pairs)—square planar shape [attached_file:1]

F. sp^3d^3 Hybridization

Mixing: 1 s orbital + 3 p orbitals + 3 d orbitals → **7 sp^3d^3 hybrid orbitals** [attached_file:1]

Characteristics:

- **Pentagonal bipyramidal geometry**

Example:

- IF_7 (iodine heptafluoride) [attached_file:1]

22.3 Summary Table of Hybridization Types

Hybridization	Orbitals Mixed	Hybrid Orbitals	Geometry	Bond Angle	Examples
sp	1s + 1p	2	Linear	180°	$BeCl_2$, C_2H_2
sp^2	1s + 2p	3	Trigonal planar	120°	BF_3, C_2H_4
sp^3	1s + 3p	4	Tetrahedral	109.5°	CH_4, NH_3, H_2O
sp^3d	1s + 3p + 1d	5	Trigonal bipyramidal	90°, 120°, 180°	PCl_5
sp^3d^2	1s + 3p + 2d	6	Octahedral	90°, 180°	SF_6
sp^3d^3	1s + 3p + 3d	7	Pentagonal bipyramidal	72°, 90°, 180°	IF_7

22.4 Quick Method to Determine Hybridization

Steric number (SN) = Number of atoms bonded to central atom + Number of lone pairs on central atom

- **SN = 2** → **sp** (linear)
- **SN = 3** → **sp²** (trigonal planar)
- **SN = 4** → **sp³** (tetrahedral)
- **SN = 5** → **sp³d** (trigonal bipyramidal)
- **SN = 6** → **sp³d²** (octahedral) [62]

Note: Multiple bonds count as **one bonding region** [62].

23. VSEPR Theory

Valence Shell Electron Pair Repulsion (VSEPR) Theory predicts molecular geometry based on the principle that **electron pairs around a central atom repel each other** and arrange themselves to **minimize repulsion** [64] [65] [66] [67] [68].

23.1 Basic Principles

1. Electron pair repulsion:
Electron pairs (both bonding and lone pairs) repel each other and adopt positions as far apart as possible [64] [65] [66] [67].

2. Order of repulsion strength:

This means:

- Lone pairs repel more strongly than bonding pairs
- Lone pairs occupy more space than bonding pairs
- Lone pairs cause deviations from ideal bond angles [64] [65] [66] [67]

3. Multiple bonds:
Double and triple bonds are treated as **single regions of electron density** for determining geometry [66] [68].

23.2 VSEPR Procedure

Step 1: Draw the Lewis structure of the molecule

Step 2: Count the total number of electron pairs around the central atom (bonding + lone pairs)

Step 3: Determine the electron pair geometry (arrangement that minimizes repulsion)

Step 4: Identify the molecular geometry based on the positions of **atoms only** (ignore lone pairs when naming molecular shape)

Step 5: Predict deviations from ideal bond angles due to lone pair repulsion [64] [66] [68]

23.3 Common Geometries

A. Two electron pairs (SN = 2):

- **Electron pair geometry:** Linear
- **Molecular geometry:** Linear
- **Bond angle:** 180°
- **Example:** CO_2, $BeCl_2$ [66] [68]

B. Three electron pairs (SN = 3):

Bonding pairs	Lone pairs	Molecular geometry	Bond angle	Example
3	0	Trigonal planar	120°	BF_3
2	1	Bent (angular)	<120° (~118°)	$SnCl_2$

50][56

C. Four electron pairs (SN = 4):

Bonding pairs	Lone pairs	Molecular geometry	Bond angle	Example
4	0	Tetrahedral	109.5°	CH_4
3	1	Trigonal pyramidal	<109.5° (~107°)	NH_3
2	2	Bent (angular)	<109.5° (~104.5°)	H_2O

47][^50][56

D. Five electron pairs (SN = 5):

Bonding pairs	Lone pairs	Molecular geometry	Example
5	0	Trigonal bipyramidal	PCl_5
4	1	Seesaw (distorted tetrahedral)	SF_4
3	2	T-shaped	ClF_3
2	3	Linear	XeF_2

Note: Lone pairs preferentially occupy **equatorial positions** in trigonal bipyramidal geometry because these positions have more space (120° angles) compared to axial positions (90° angles) [64] [65] [66] [attached_file:1].

E. Six electron pairs (SN = 6):

Bonding pairs	Lone pairs	Molecular geometry	Example
6	0	Octahedral	SF_6
5	1	Square pyramidal	BrF_5
4	2	Square planar	XeF_4

Note: In octahedral geometry, all positions are equivalent. When there are two lone pairs, they occupy **opposite positions** (180° apart) to minimize repulsion [65] [66][attached_file:1].

23.4 Effect of Lone Pairs on Bond Angles

Lone pairs cause bond angles to be **smaller than ideal values** because [64] [65] [67]:

- Lone pairs occupy more space than bonding pairs
- Greater lone pair–bond pair repulsion pushes bonding pairs closer together

Examples:

- **CH_4:** No lone pairs → bond angle = 109.5° (ideal tetrahedral)
- **NH_3:** One lone pair → bond angle ≈ 107° (slightly compressed)
- **H_2O:** Two lone pairs → bond angle ≈ 104.5° (more compressed) [65] [68][attached_file:1]

24. Geometry of Molecules Containing Lone Pair and Bond Pair

24.1 Ammonia (NH_3)

- **Central atom:** N
- **Bonding pairs:** 3 (three N–H bonds)
- **Lone pairs:** 1
- **Steric number:** 4
- **Electron pair geometry:** Tetrahedral
- **Molecular geometry: Trigonal pyramidal**
- **Bond angle:** ~107° (< 109.5° due to lone pair repulsion)
- **Hybridization:** sp^3 [65] [68][attached_file:1]

24.2 Water (H_2O)

- **Central atom:** O
- **Bonding pairs:** 2 (two O–H bonds)
- **Lone pairs:** 2
- **Steric number:** 4
- **Electron pair geometry:** Tetrahedral
- **Molecular geometry: Bent (angular)**
- **Bond angle:** ~104.5° (< 109.5° due to two lone pairs)
- **Hybridization:** sp^3 [65] [68][attached_file:1]

24.3 Sulfur Tetrafluoride (SF_4)

- **Central atom:** S
- **Bonding pairs:** 4
- **Lone pairs:** 1
- **Steric number:** 5
- **Electron pair geometry:** Trigonal bipyramidal
- **Molecular geometry: Seesaw**
- **Lone pair position:** Equatorial (to minimize repulsion)
- **Hybridization:** sp^3d [attached_file:1]

24.4 Xenon Difluoride (XeF_2)

- **Central atom:** Xe
- **Bonding pairs:** 2
- **Lone pairs:** 3
- **Steric number:** 5
- **Electron pair geometry:** Trigonal bipyramidal
- **Molecular geometry: Linear**
- **Lone pair positions:** All three occupy equatorial positions
- **Bond angle:** 180°
- **Hybridization:** sp^3d [attached_file:1]

24.5 Xenon Tetrafluoride (XeF_4)

- **Central atom:** Xe
- **Bonding pairs:** 4
- **Lone pairs:** 2
- **Steric number:** 6
- **Electron pair geometry:** Octahedral
- **Molecular geometry: Square planar**
- **Lone pair positions:** Opposite to each other (180° apart)
- **Bond angle:** 90°, 180°
- **Hybridization:** sp^3d^2 [65] [attached_file:1]

25. Metallic Bonding

Metallic bonding is the type of chemical bonding responsible for holding metal atoms together in a metallic solid. It explains the unique properties of metals such as electrical conductivity, malleability, ductility, and luster [11] [16] [20] [69] [70].

25.1 Nature of Metallic Bonding

In metallic bonding:

- Metal atoms lose their valence electrons, which become **delocalized**
- These free electrons form an "**electron sea**" or "**electron gas**" that surrounds positively charged metal ions (cations or kernels)
- The **electrostatic attraction** between the delocalized electrons and positive metal ions constitutes the metallic bond [11] [16] [69] [70] [attached_file:1]

Key features:

- Metallic bonds are **non-directional** (unlike covalent bonds)
- Electrons are **mobile** and not localized between specific atoms
- The bonding extends throughout the entire metal structure [11] [16] [69]

25.2 Properties Explained by Metallic Bonding

1. Electrical conductivity:

- Free electrons can move under an applied electric field
- Metals are excellent conductors of electricity [69] [70] [attached_file:1]

2. Thermal conductivity:

- Free electrons transfer kinetic energy rapidly through the metal
- Metals efficiently conduct heat [69] [attached_file:1]

3. Malleability and ductility:

- Metal ions can slide past each other without breaking bonds (non-directional bonding)
- The electron sea adjusts to maintain bonding in new positions
- Metals can be hammered into sheets (malleable) or drawn into wires (ductile) [11] [16] [69] [70]

4. Luster (shine):

- Free electrons can absorb and re-emit light of various wavelengths
- Metals have a characteristic shiny appearance [69] [attached_file:1]

5. High melting and boiling points:

- Strong electrostatic attraction between metal ions and electron sea
- Significant energy required to overcome metallic bonding [attached_file:1]

26. The Classical Free Electron Theory

The classical free electron theory (also called the **Drude-Lorentz model**) was the first attempt to explain metallic properties using the concept of free electrons [71] [72] [73] [74] [attached_file:1].

26.1 Key Assumptions

1. Free electron gas:

- Valence electrons are completely free to move throughout the metal
- These electrons behave like molecules in an ideal gas [71] [72] [73] [74]

2. Potential energy:

- Electrons inside the metal have lower potential energy than electrons outside
- A potential barrier at the surface prevents electrons from escaping [74]

3. Independent motion:

- Electrons move independently of each other
- Collisions with metal ions cause resistance [72] [73]

26.2 Successes of the Classical Theory

- Explained **electrical conductivity** qualitatively
- Explained **thermal conductivity** (Wiedemann-Franz law)
- Accounted for **Ohm's law** [72] [73] [attached_file:1]

26.3 Failures and Limitations

- Could not explain the **low electronic specific heat** of metals
- Failed to explain temperature dependence of resistivity accurately
- Could not account for differences in conductivity among different metals
- Did not explain **magnetic properties** of materials [72] [73] [74] [attached_file:1]

Note: The failures led to the development of quantum mechanical models (band theory).

27. The Valence Bond Theory (for Metals)

The valence bond approach to metallic bonding treats the cohesion as arising from **resonance among many equivalent metal-metal covalent bonds** [attached_file:1].

- This model suggests that metallic bonding is an extension of covalent bonding
- The delocalization of bonding electrons results from resonance structures
- While providing some insight, this model is limited and has been largely superseded by band theory [attached_file:1]

28. The Band Theory (or Band Theory of Metals)

Band theory is a **quantum mechanical model** that successfully explains the electronic properties of solids, including metals, semiconductors, and insulators [69] [72] [70] [73] [75].

28.1 Basic Concept

When a large number of atoms (N) come together to form a solid:

- Atomic orbitals from all atoms overlap
- Form **molecular orbitals** (by LCAO principle)
- The number of molecular orbitals equals the number of atomic orbitals (N orbitals per atom × N atoms)
- These closely spaced energy levels merge into **continuous energy bands** [69] [70] [73] [75] [attached_file:1]

28.2 Types of Bands

1. Valence band:

- Formed from valence atomic orbitals
- Contains the outermost electrons of atoms
- In metals at 0 K, this band is **partially filled** or overlaps with conduction band [69] [70] [75]

2. Conduction band:

- Higher energy band
- In metals, this band **overlaps with or is partially filled** by valence band electrons
- Electrons in this band are free to move and conduct electricity [69] [70] [75]

3. Band gap (or forbidden gap):

- Energy difference between valence band and conduction band
- Determines whether a material is a conductor, semiconductor, or insulator [70] [75]

28.3 Classification Based on Band Theory

A. Conductors (Metals):

- **Valence band and conduction band overlap** or valence band is partially filled
- **No band gap** (or very small gap)
- Electrons can easily move from valence to conduction band
- **High electrical conductivity** [69] [70] [75]

Examples: Cu, Ag, Au, Na, Al

B. Semiconductors:

- **Small band gap** (typically 0.1 to 3 eV)
- At 0 K, valence band is full; conduction band is empty
- At higher temperatures or with doping, some electrons can jump to conduction band
- **Moderate electrical conductivity** that increases with temperature [70] [75]

Examples: Si, Ge

C. Insulators:

- **Large band gap** (> 3 eV)
- Valence band is completely filled; conduction band is empty
- Electrons cannot easily jump to conduction band
- **Very low or no electrical conductivity** [70] [75]

Examples: Diamond, glass, rubber

28.4 Fermi Level

The **Fermi level (EF)** is the highest energy level occupied by electrons at **absolute zero temperature (0 K)** [70] [73].

- In metals, the Fermi level lies **within the conduction band**
- The Fermi surface describes the distribution of electron energies [70] [73]

28.5 Advantages of Band Theory

- Explains conductivity differences among conductors, semiconductors, and insulators
- Accounts for temperature dependence of resistivity
- Explains optical, magnetic, and thermal properties of solids
- Forms the basis for understanding semiconductor devices [69] [70] [75]

29. Secondary Bond Forces and All Interactions

Secondary bond forces (also called **intermolecular forces**) are weak electrostatic attractions between molecules. They are much weaker than primary bonds (ionic, covalent, metallic) but play crucial roles in determining physical properties like boiling point, melting point, and solubility [76] [77] [78] [79] [80] [81] [82] [83] [84].

29.1 Types of Intermolecular Forces

A. Ion-Dipole Interactions

Description:

- Attractive force between an **ion** and a **polar molecule** (dipole)
- One of the strongest intermolecular forces [81] [82] [83] [84]

Importance:

- Responsible for **dissolution of ionic compounds** in polar solvents like water
- Cation attracts the negative end of water dipole; anion attracts the positive end
- This is the basis of **hydration** and **solvation energy** [82] [83] [84]

Example: Na^+ and Cl^- ions dissolving in water

B. Dipole-Dipole Interactions

Description:

- Attractive force between the **positive end of one polar molecule** and the **negative end of another polar molecule**
- Molecules orient to maximize attraction and minimize repulsion [85] [80] [81] [82] [83] [84]

Strength:

- Weaker than ion-dipole but stronger than dispersion forces
- Strength is proportional to dipole moment and inversely proportional to distance cubed ($\propto 1/r^3$) [82] [83]

Examples: HCl···HCl, acetone···acetone

C. Dipole-Induced Dipole Interactions

Description:

- A **permanent dipole** induces a **temporary dipole** in a neighboring **nonpolar molecule** or atom
- The induced dipole then interacts with the permanent dipole [80] [81] [attached_file:1]

Example: HCl molecule near an O_2 molecule

D. London Dispersion Forces (Induced Dipole-Induced Dipole)

Description:

- Weak attractive forces arising from **temporary fluctuations in electron density**
- Create instantaneous dipoles that induce dipoles in neighboring molecules
- Present in **all molecules** (polar and nonpolar) [85] [80] [81] [82] [83] [84]

Factors affecting strength:

- **Size/molecular mass:** Larger molecules/atoms have more electrons → stronger dispersion forces
- **Surface area:** Greater contact area → stronger dispersion forces [80] [81] [82]

Examples: Noble gases (He, Ne, Ar), nonpolar molecules (CH_4, Cl_2)

Note: London forces are the **weakest** intermolecular forces but become significant in large molecules [80] [81] [82] [84].

29.2 Comparison of Intermolecular Forces

Force	Strength (kJ/mol)	Nature	Examples
Ion-dipole	40-600	Ion + polar molecule	NaCl in water
Hydrogen bonding	10-40	H bonded to N/O/F + lone pair	H_2O, NH_3, DNA
Dipole-dipole	5-25	Polar molecule + polar molecule	HCl···HCl
London dispersion	0.05-40	All molecules (temporary dipoles)	CH_4, noble gases

[94][97]

For comparison:

- Covalent bond: 150-1000 kJ/mol
- Ionic bond: 400-4000 kJ/mol
- Intermolecular forces: < 40 kJ/mol [84]

29.3 Effect on Physical Properties

1. Boiling and melting points:

- Stronger intermolecular forces → higher boiling/melting points
- Substances must overcome intermolecular forces to change phase [80] [82] [84]

2. Solubility:

- "Like dissolves like"—polar substances dissolve in polar solvents; nonpolar in nonpolar
- Determined by the balance of intermolecular forces [82] [83]

3. Viscosity and surface tension:

- Stronger intermolecular forces → higher viscosity and surface tension [86]

30. Hydrogen Bonding and Types

Hydrogen bonding is a special type of **strong dipole-dipole interaction** that occurs when hydrogen is bonded to highly electronegative atoms (F, O, N) and interacts with a lone pair on another electronegative atom [76] [77] [78] [79] [86] [82] [83] [84].

30.1 Conditions for Hydrogen Bonding

1. Hydrogen donor:

- Hydrogen atom **covalently bonded** to F, O, or N [76] [77] [78] [82] [83]

2. Hydrogen acceptor:

- Another electronegative atom (F, O, N) with a **lone pair of electrons** [76] [77] [78] [82] [83]

3. Strong electronegativity difference:

- Creates sufficient partial positive charge (δ^+) on hydrogen [76] [78] [82]

30.2 Nature of Hydrogen Bonding

- **Strength:** 10-40 kJ/mol (stronger than typical dipole-dipole but weaker than covalent/ionic bonds) [76] [82] [83] [84]
- **Directionality:** Hydrogen bonds have directional character [76] [78]
- **Distance:** H-bond distances are shorter than typical van der Waals distances [76]

Representation: Usually shown as dotted or dashed lines (X–H···Y) [76] [86]

30.3 Types of Hydrogen Bonding

A. Intermolecular Hydrogen Bonding

Definition:
Hydrogen bonding that occurs **between different molecules** [87] [76] [77] [78] [79] [86].

Characteristics:

- Causes **association** of molecules
- **Increases boiling point, melting point, and viscosity**
- Responsible for many unique properties of substances [76] [78] [86]

Examples:

1. Water (H_2O):

- Each water molecule can form up to **4 hydrogen bonds** (2 as donor, 2 as acceptor)
- Explains water's unusually high boiling point (100°C) compared to H_2S (-60°C)
- Explains ice being **less dense than liquid water** (open tetrahedral structure in ice) [76] [78] [86] [82]

2. Ammonia (NH_3):

- N–H···N hydrogen bonding between molecules
- Higher boiling point than expected [76] [86]

3. Hydrogen fluoride (HF):

- F–H···F hydrogen bonding
- Extremely strong due to high electronegativity of fluorine
- Forms chains and clusters [76] [86]

4. Alcohols and carboxylic acids:

- O–H···O hydrogen bonding
- Higher boiling points than corresponding hydrocarbons of similar mass [76] [86]

5. DNA double helix:

- Adenine-thymine (A-T): 2 hydrogen bonds
- Guanine-cytosine (G-C): 3 hydrogen bonds
- Stabilizes the double helix structure [77] [78]

6. Proteins:

- N–H···O=C hydrogen bonding
- Maintains secondary structures (α-helix, β-sheet) and tertiary structure [77] [78]

B. Intramolecular Hydrogen Bonding

Definition:
Hydrogen bonding that occurs **within the same molecule** [87] [76] [77] [78] [79] [86].

Characteristics:

- Stabilizes certain molecular conformations
- **Decreases boiling point** (molecules are less associated)
- Increases **volatility** and **solubility in nonpolar solvents** [86] [attached_file:1]

Examples:

1. o-Nitrophenol:

- Hydrogen bonding between –OH and $–NO_2$ groups within the same molecule
- Lower boiling point than p-nitrophenol (which has intermolecular H-bonding)
- More volatile, steam-distillable [86] [attached_file:1]

2. Salicylic acid:

- Intramolecular H-bonding between –COOH and –OH groups [86]

3. Ethylene glycol ($C_2H_4(OH)_2$):

- Can form intramolecular H-bond between the two –OH groups when they are close [76]

30.4 Consequences of Hydrogen Bonding

1. Abnormal boiling and melting points:

- H_2O, HF, NH_3 have much higher boiling points than expected from their molecular masses [76] [86]

2. Solubility:

- Alcohols, sugars, amino acids are soluble in water due to H-bonding with water [86]

3. Lower density of ice:

- Ice has an open hexagonal structure with H-bonds
- Density of ice < density of liquid water
- Ice floats on water [76] [86]

4. Viscosity:

- Liquids with extensive H-bonding (glycerol, honey) are very viscous [86]

5. Biological significance:

- DNA base pairing
- Protein folding and stability
- Enzyme-substrate recognition [77] [78]

Summary: Quick Reference Guide for Midterm

Key Formulas

1. Bond order:

$$\text{BO} = \frac{1}{2}(N_b - N_a)$$

2. Dipole moment:

$$\mu = q \times d$$

3. Percent ionic character:

$$\%\ \text{ionic} = \left(\frac{\mu_{\text{obs}}}{\mu_{\text{ionic}}}\right) \times 100$$

4. Born-Haber cycle (general form):

$$\Delta H_f^\circ = \Delta H_{at} + \frac{1}{2}D + \sum IE + \sum EA + U$$

5. Enthalpy of solution:

$$\Delta H_{sol} = \Delta H_{lattice} + \Delta H_{hydration}$$

6. Steric number:

$$\text{SN} = (\text{bonding pairs}) + (\text{lone pairs})$$

Important Concepts to Remember

1. **Quantum numbers:** n (size), l (shape), m (orientation), s (spin)
2. **Electronic configuration rules:** Aufbau, Pauli, Hund
3. **Bond types:** Ionic (electron transfer), covalent (electron sharing), metallic (electron sea)
4. **VBT vs. MOT:** VBT uses orbital overlap; MOT forms bonding/antibonding MOs
5. **Fajan's rules:** Small cation + large anion → more covalent character
6. **VSEPR:** Electron pairs repel; lp-lp > lp-bp > bp-bp

7. **Hybridization:** Predicts geometry; SN determines hybrid type
8. **H-bonding:** Strong dipole-dipole with H bonded to F/O/N
9. **Band theory:** Explains conductors (no gap), semiconductors (small gap), insulators (large gap)

Common Exam Questions Topics

- Calculate lattice energy using Born-Haber cycle
- Predict molecular geometry using VSEPR
- Determine hybridization from molecular structure
- Draw MO diagrams and calculate bond order
- Explain trends using Fajan's rules
- Compare intermolecular forces and predict boiling points
- Identify types of hydrogen bonding

Best of luck on your midterm exam!
[88] [89] [90] [91] [92] [93] [94] [95] [96] [97] [98] [99] [100]

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